PORTFOLIO · 2026

Wash Log

A phone photo of the windscreen becomes a checked row in the spreadsheet management already reads.



Givi Jananashvili · Product owner & developer

Internal web app for a premium automotive service centre · 3 car washers logged

THE PROBLEM

A hand-typed log nobody could trust

One car in thirty was wrong

Of 1,757 rows, 60 VINs could not exist: wrong length, letters a VIN never uses, typos.

Excel filled in the future

A run of dates auto-completed into 2027–2036. The sheet looked fine; the months did not add up.

The sheet is the report

Management reads and corrects the spreadsheet directly. Replacing it with a database was not an option.

HOW IT WORKS

Photos in, rows out

01

Shift

Pick washer and day

Decides the sheet and the date. The date is stored as text, exactly as the sheet always had it.

02

Photos

Drop the day's photos

Straight from the messenger, or paste. On a phone, the built-in text scanner fills the field instead.

03

Check

Every VIN colour-coded

Green is fine, amber means look again, red cannot be written.

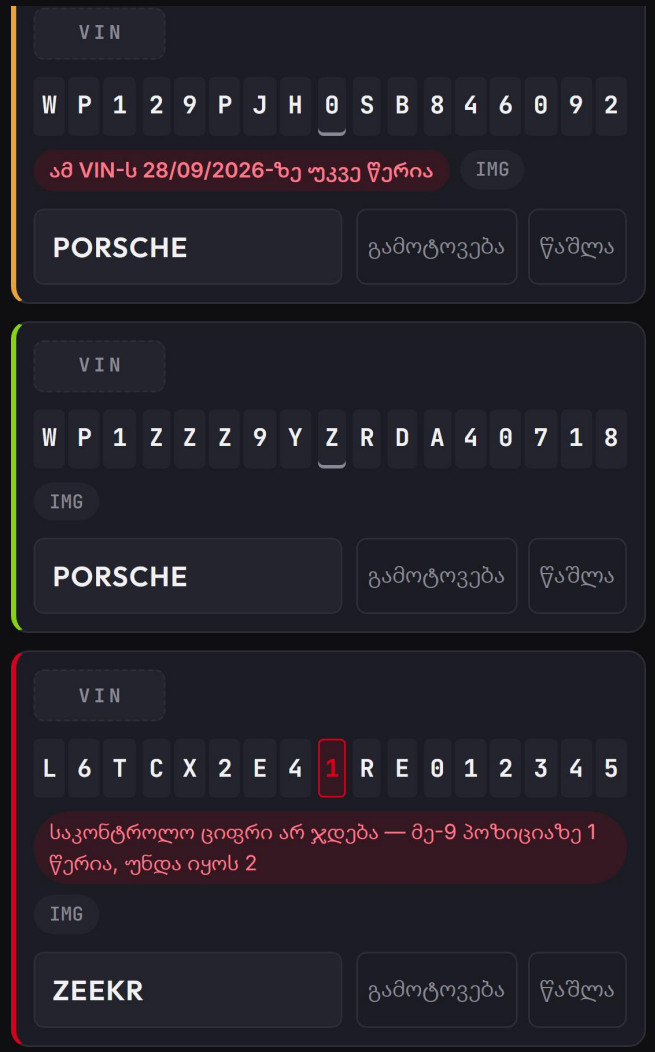
04

Write

Rows land in the sheet

Same columns, same format, one click. The spreadsheet stays the original.

Photos are read and discarded. Only the spreadsheet is kept.



Sample data

WHAT IT CHECKS

Wrong before it is written

Structure

17 characters from the VIN alphabet. The letters I, O and Q never occur, so OCR's O becomes 0 automatically.

Check digit, where it is real

Position 9 is verified only for manufacturers whose VINs actually carry one. The next slide is why.

Duplicates and strangers

The same VIN on the same day is flagged, and so is a manufacturer code the log has never seen.

THE KEY DECISION

The rule came from the data, not from the standard.

7–13 %

of one maker's VINs pass: chance

99–100 %

for the other makers in the log

European VINs may skip the check digit. Enforcing it everywhere would have rejected a whole brand. The app enforces it per manufacturer code only when the log proves it applies: at least 5 VINs, at least 85 % passing.

WHAT MANAGEMENT READS

Washes per person, per month



This month, at a glance

One tile per washer with the change against last month.

Every wash counts

A car washed three times is three washes. A deliberate choice: the figure is work done, not the size of the fleet.

Counted on the way in

The same pass that loads the sheet also counts it, so the figures always match the file.

Sample data · made-up names

HOW IT IS BUILT

The architecture

Phones

Installable web app with a service worker. The phone's own scanner reads the VIN in two seconds.

Google Cloud Run

Python, FastAPI and a local OCR model (ONNX). Scales to zero, so it costs nothing overnight.

Private dataset repo

The .xlsx itself. Every write is a commit; a write based on a stale copy is rejected, not merged.

5.6 s

per photo with local OCR, no paid API

< 1 s

app start once the container is awake

195 s

from a push to main to live

HOW IT IS BUILT

Reliability and security

Fail-closed by default

Hosted without a password configured, the app refuses every request rather than opening up.

Lockout

Eight wrong attempts lock that address for five minutes. Changing the password ends every session.

Nothing kept that is not needed

Photos are read in memory and dropped. Only the spreadsheet is stored.

No silent overwrites

Two people writing at once: the second write is refused instead of silently replacing the first.

Opens with no signal

The app shell loads from the phone's cache in milliseconds, whatever the server is doing.

A login page is not a stylesheet

An expired session redirects every file to login. The cache refuses redirected responses, so it never stores the wrong page.

WHAT I LEARNED

Lessons from shipping it

Free hosting has fine print

The first host's free hours were per account, not per service: suspended after one day. Its tiny CPU took 60 s per photo. Measure before you move.

The laptop you cannot install on

The office laptop needed admin rights for everything. The answer was the phone: a browser, a password and the scanner it already has.

A space before a variable name

Two hours lost to a leading space in a setting. Now a deliberate wrong password tells the two cases apart: 500 means unset, 401 means working.

SUMMARY

The project in numbers

60

broken VINs found in the old log

3

washers, one shared log

0

installs: a link and a password

195 s

push to live

Python · FastAPI · RapidOCR (ONNX) · openpyxl · installable web app · Google Cloud Run

givijananashvili40@gmail.com · linkedin.com/in/givi-jananashvili-a9877026b · github.com/Azazell